

Network Address Translation Configuration

College of Science, Engineering, and Technology, Grand Canyon University

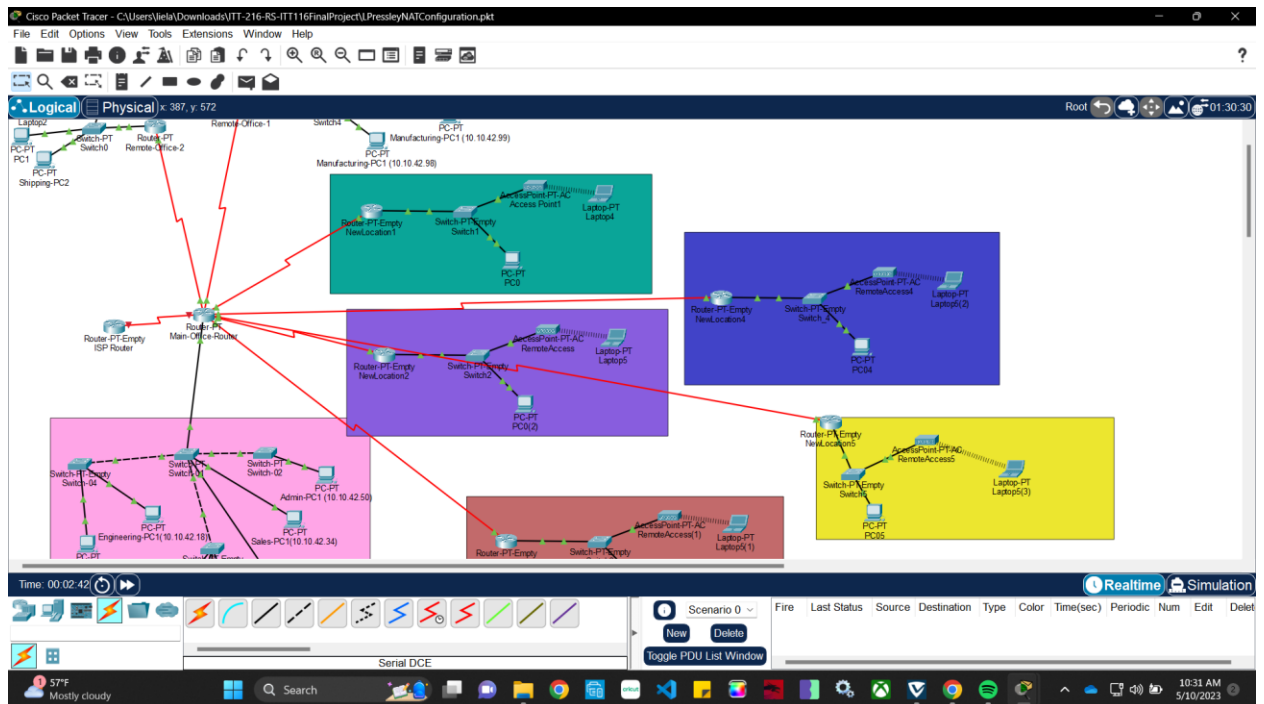
ITT-216: Enterprise Route & Switch

Instructor Tim Collins

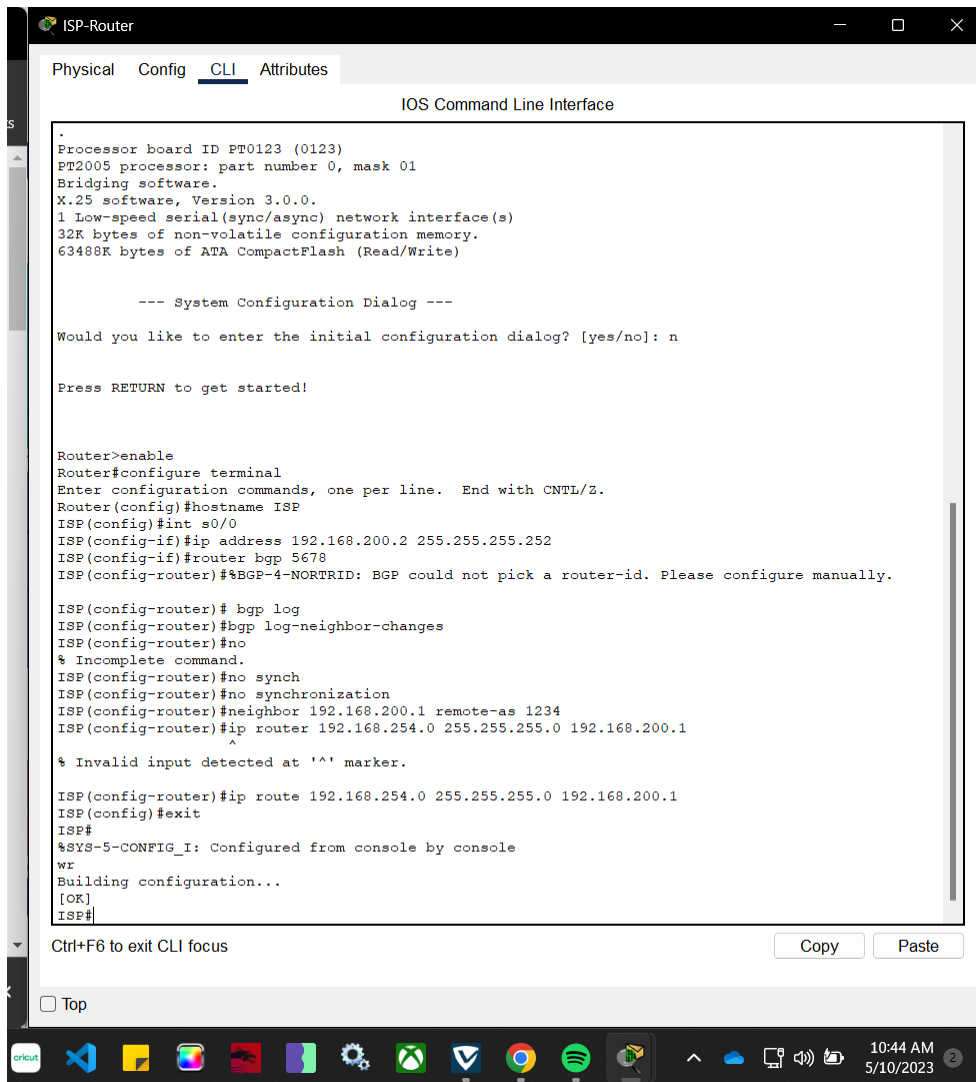
Liela Pressley

5/10/2023

1. Created and connected an ISP router to the Main Router via Serial Connection



2. Configure bgp, neighbor, and ip routes



The screenshot shows a terminal window titled "ISP-Router" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The terminal output shows the following sequence of commands and responses:

```
.
Processor board ID FT0123 (0123)
PT2005 processor: part number 0, mask 01
Bridging software.
X.25 software, Version 3.0.0.
1 Low-speed serial(sync/async) network interface(s)
32K bytes of non-volatile configuration memory.
63488K bytes of ATA CompactFlash (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: n

Press RETURN to get started!

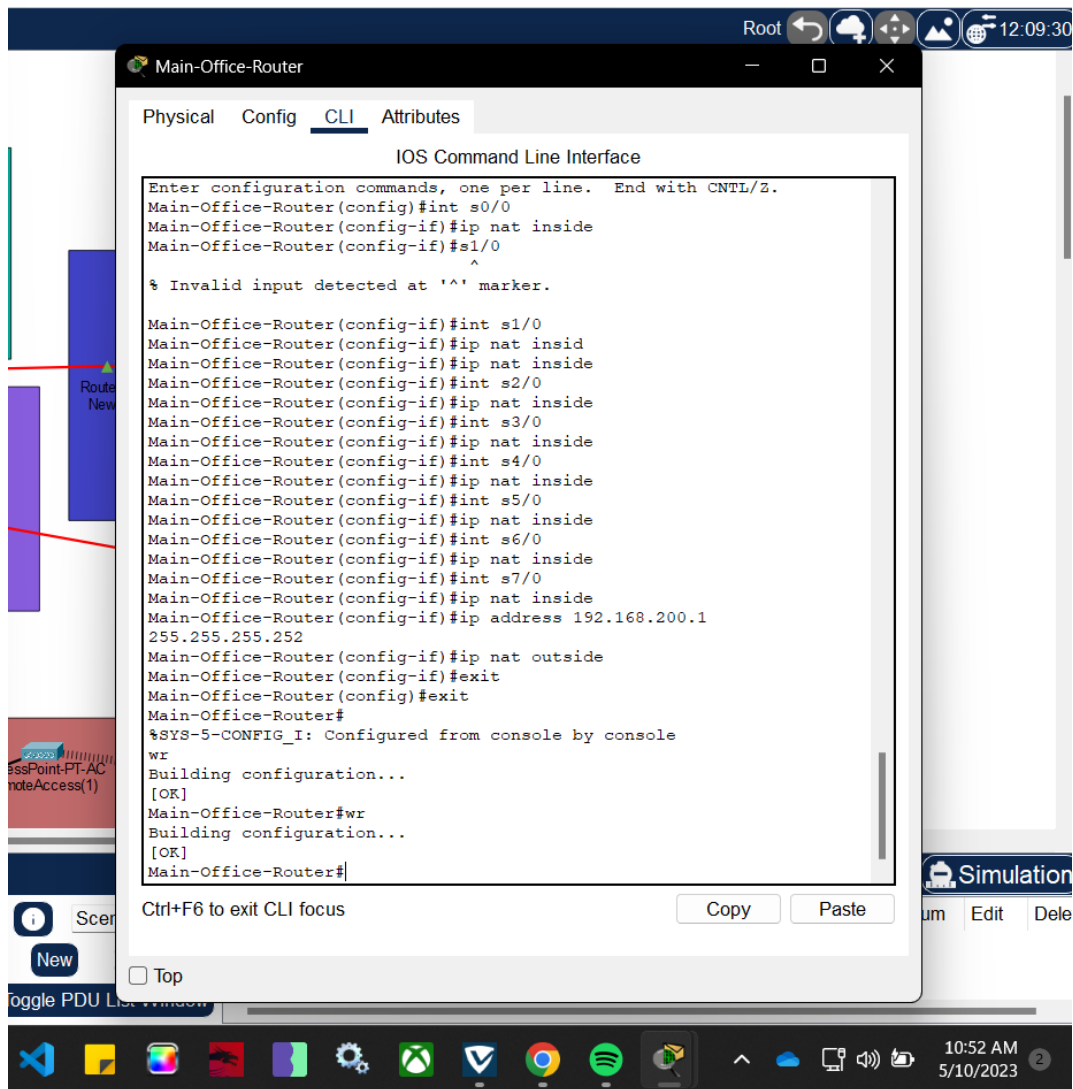
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ISP
ISP(config)#int s0/0
ISP(config-if)#ip address 192.168.200.2 255.255.255.252
ISP(config-if)#router bgp 5678
ISP(config-router)#%BGP-4-NORTRID: BGP could not pick a router-id. Please configure manually.

ISP(config-router)# bgp log
ISP(config-router)#bgp log-neighbor-changes
ISP(config-router)#no
% Incomplete command.
ISP(config-router)#no synch
ISP(config-router)#no synchronization
ISP(config-router)#neighbor 192.168.200.1 remote-as 1234
ISP(config-router)#ip router 192.168.254.0 255.255.255.0 192.168.200.1
^
% Invalid input detected at '^' marker.

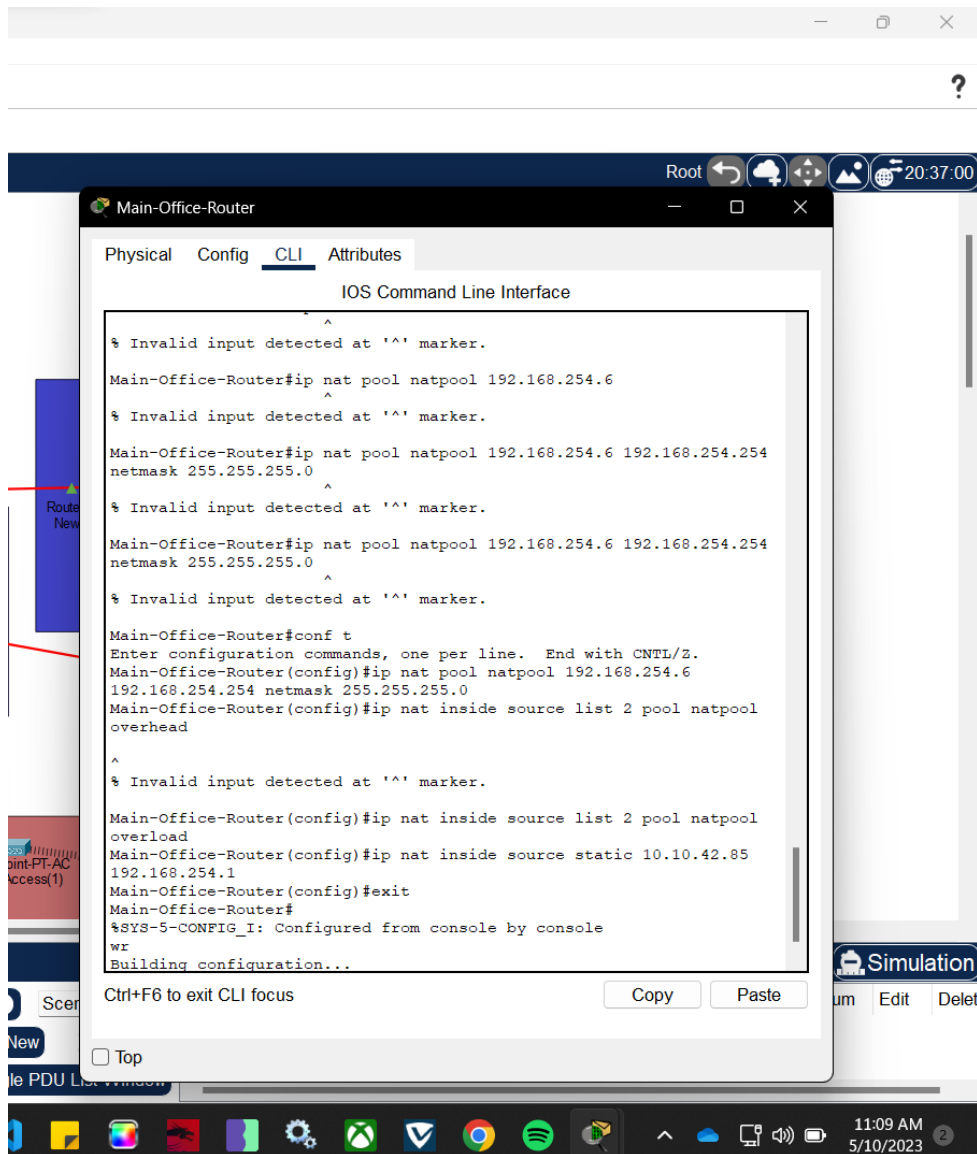
ISP(config-router)#ip route 192.168.254.0 255.255.255.0 192.168.200.1
ISP(config)#exit
ISP#
%SYS-5-CONFIG_I: Configured from console by console
wz
Building configuration...
[OK]
ISP#
```

Below the terminal window, there are "Copy" and "Paste" buttons, and a "Ctrl+F6 to exit CLI focus" message. At the bottom of the window, there is a "Top" button and a taskbar with various application icons and a system clock showing 10:44 AM on 5/10/2023.

3. Configuring NAT for each serial port on main office router



4. Setting up the NAT pool on the main office router



5. Can view inside and outside NAT translations

The screenshot shows a network simulation environment with a 'Main-Office-Router' window open. The 'CLI' tab is selected, displaying the 'IOS Command Line Interface'. The terminal shows the following commands and output:

```
overload
Main-Office-Router(config)#ip nat inside source static 10.10.42.85
192.168.254.1
Main-Office-Router(config)#exit
Main-Office-Router#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
Main-Office-Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Main-Office-Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.254.1
Main-Office-Router(config)#exit
Main-Office-Router#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
Main-Office-Router#wr
Building configuration...
[OK]
Main-Office-Router#sh ip nat translations
Pro Inside global      Inside local      Outside local      Outside
global
--- 192.168.254.1      10.10.42.85      ---                ---

Main-Office-Router#
Main-Office-Router#
Main-Office-Router#
Main-Office-Router#
Main-Office-Router#sh ip nat translations
Pro Inside global      Inside local      Outside local      Outside
global
--- 192.168.254.1      10.10.42.85      ---                ---

Main-Office-Router#
```

Below the terminal output, there is a 'Ctrl+F6 to exit CLI focus' message and 'Copy' and 'Paste' buttons. The background shows a network diagram with a 'Router New' label and a 'AccessPoint-PT-AC RemoteAccess(1)' label. The bottom of the screen displays a Windows taskbar with various application icons and a system clock showing 11:13 AM on 5/10/2023.